

Gentle by Knowing: How Much Must a Care Robot Know About a Person to Lift Their Arm Safely?

Anonymous authors

Abstract—Robots that move a person’s limb are protected today mainly by limits on contact force. We asked how much a robot must know about the specific person to lift their forearm gently, that is, without pressing a joint into the end of its range of motion, and whether force sensing can take the place of that knowledge. In a musculoskeletal simulation of a robot supporting the forearm from below, in a simulated care population with restricted shoulders and elbows, we compared controllers that differ only in what they know about the person. No force-threshold guard reduced harmful outcomes by significantly more than 5 percentage points; its best setting cut harm by 4–7 points at the cost of 40–51% of safe completions, because 95–100% of harmful outcomes began during the first attempt. A decision policy learned offline in simulation reduced harmful outcomes from 110 to 25 of 300 persons at a similar number of safe completions (174 vs. 177), using a care record with 20° error. Knowledge helped mainly by telling the robot whom not to lift.

I. INTRODUCTION

Moving a person’s limb is among the most frequent actions in care, and a natural target for assistive robots [1]–[3]. Doing it gently is harder than it looks, because many care recipients cannot move freely. About half of adults admitted with stroke develop at least one contracture within six months [4], and contractures are common among residents of long-term care [5]. In long-term care, contractures independently predict minimal-trauma fractures [6], some of which occur during diaper changes, repositioning or transfers [7]; in hemiplegia, the way the arm is moved is associated with shoulder pain [8], [9].

Collaborative-robot safety rests largely on limits on contact force and pressure derived from pain-onset thresholds [10]. Standards in adjacent scopes ask more: wearable assistive robots may apply assist force only within the user’s passive range of motion [11], rehabilitation robots must limit movement to pre-set, patient-specific limits [12], and personal-care robots are expected to seek external assistance or stop when decision uncertainty remains unacceptable [13]. For a robot that moves a limb from the outside, by end-effector contact, it is not known which of these measures matter, nor how much the robot needs to know about the particular person.

Is a sensitive force limit enough, or does the robot need a detailed personal model of the body? Human carers suggest a third answer: they read the chart, notice that “the left shoulder is stiff”, and handle the arm differently or ask for help. We make the robot’s knowledge of the person the independent variable, and measure gentleness as the absence of load pressed into the end of a joint’s range.

Contributions. (1) An *information ladder* for robot limb handling: harmful outcomes and safe completions at seven

levels of knowledge about the person (Table I, Fig. 3). (2) Evidence that force-threshold control does not substitute for this knowledge in end-effector support of a multi-joint limb, with the mechanism: harm begins in the first attempt (Fig. 2). (3) A decision policy learned offline in simulation that reduces harmful outcomes to about a quarter of a fixed lift’s at a similar number of safe completions, from a coarse care record; it replicates and survives several degradations, but loses much of its benefit when the record misses restrictions (Fig. 4). (4) A post hoc analysis of what the robot needs to know: which joint is restricted, not by how many degrees.

II. RELATED WORK

Robots already move passive limbs within joint limits: Chow and Kemp [1] predicted limb motion under robot support with joint-limit constraints, and Manip4Care repositions limbs within age-group joint limits and pauses when no valid pose exists [3]. TOORAD plans dressing within each participant’s measured range [14], GRACE predicts functional range from clinical scores [15], assistive policies have been trained for simulated impairments [16], and musculoskeletal models have been placed inside the planner for shoulder rehabilitation [17], [18]. None reports harm across levels of information about the person, declines when no support is predicted safe, or compares against force-threshold control. Comparing a baseline, an inference model and an omniscient planner on a safety–efficiency plane has precedent in autonomous driving [19], and generic and personalised human models have been compared against realism, tracking or reward [20]–[22]. The same external force can correspond to very different joint loads [23], and industrial force limits do not transfer directly to patients [24]. In joint-aligned single-degree-of-freedom devices, torque thresholds are used to find the end of range [25]; our claim is therefore scoped to end-effector support of a multi-joint limb, where the loaded joint is not on the sensor’s axis. Robots that ask for help under ambiguity [26] establish declining as a legitimate decision; here it is triggered by predicted harm to the body.

III. METHODS

A. Simulated body and population

We used the MyoS uite myoArm model [27] (MyoS uite 2.12.2) in MuJoCo 3.6 [28], with the humerus not welded to the thorax, so the arm hangs from the shoulder complex. Muscles were driven by a stretch reflex with muscle velocity and length feedback, following the structure of [29], with a 25-ms afferent delay (our choice) and a tonic baseline activation

proximal: shoulder +16°, elbow -3° distal: shoulder +9°, elbow +10°

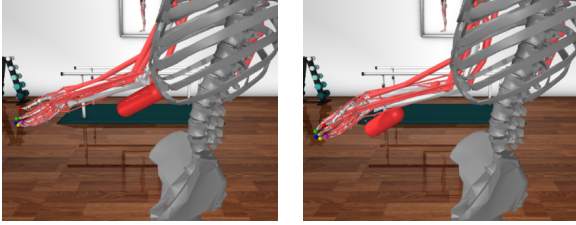


Fig. 1. The task. A capsule supports the forearm from below and raises it 5 cm. The support site decides which joint moves (joint excursions for one unrestricted simulated person, seed 104). For a person with a restricted shoulder or elbow, the wrong site presses that joint into the end of its range.

of 0.01; there is no voluntary activity. Each simulated person's muscle strength, passive muscle force and joint damping were varied log-normally with $\sigma = 0.25$, and segment masses and inertias with $\sigma = 0.175$.

Each person independently had a restricted shoulder elevation with probability 0.4 and a restricted elbow flexion with probability 0.3, following reported contracture rates [4], [5]. A restricted joint could move a free range drawn from $U(4^\circ, 16^\circ)$ (shoulder) or $U(0^\circ, 10^\circ)$ (elbow) above the person's own settled posture before its joint limit engaged. The free ranges were chosen to overlap the joint excursions produced by the task (about 7° – 14° at the shoulder and up to 9° at the elbow); absolute harm rates are therefore properties of the population design, and we interpret only differences between controllers on the same persons. A *shifted* population had restriction probabilities 0.6 (shoulder, $U(2^\circ, 12^\circ)$) and 0.5 (elbow, $U(0^\circ, 8^\circ)$). By default the joint limit is hard; in *graded end-feel* conditions, resistance ramps up over a window of width $w \in \{6^\circ, 12^\circ, 20^\circ, 25^\circ\}$ centred on the limit.

B. Robot and task

The end effector is a position-servoed capsule (effectively five degrees of freedom; stiffness 6000 N/m and 20 N-m/rad, 50-Hz command rate) that approaches the forearm from below, raises the forearm centre by 5 cm and holds for 0.5 s. A *strategy* is a support site along the forearm (8 sites, 3 cm apart), an approach angle (-40° , -20° , 0°) and a lifting speed (0.10 or 0.25 m/s), giving 48 strategies. Proximal support elevates mainly the shoulder; distal support also flexes the elbow (Fig. 1). No single strategy is safe for everyone.

C. Gentleness: harm and outcomes

Harm is the load the robot presses into the shoulder-elevation and elbow-flexion limits: the peak of the 50-ms moving average of the joint-limit constraint torque (excluding the first 100 ms), summed over the two joints. From round 23 onward, any passive limit torque present before contact was subtracted. An outcome is *harmful* if harm ≥ 0.5 N-m (primary threshold). We also report 2.5 N-m, close to the mean torque (2.7 ± 1.1 N-m) that therapists applied to reach passive elbow end range in children after fracture [30], a threshold sweep, and a threshold-free mean load. A *safe completion* is

a completed lift with harm below the threshold; a declined lift counts as neither. Harm is shown to no controller except the oracle. It is a mechanical exposure proxy, not pain or injury.

D. What the robot may know

The *care record* gives each joint's free range with Gaussian error of standard deviation $\varepsilon \in \{4^\circ, 10^\circ, 20^\circ\}$; standard errors of measurement of 14 – 25° between raters have been reported for symptomatic shoulders [31], so 20° is our realistic default. Unrestricted joints are recorded as unrestricted. Real records are often categorical, such as the MDS 3.0 upper-extremity item [32], and often incomplete: in one nursing-home study, reduced range of motion was found in 69% of residents but documented for 4% [33]. We therefore also consider a categorical limitation flag (restricted or not, shoulder and elbow merged) and a *half-documented* record in which each restricted person is unrecorded with probability 0.5. The robot observes the person's settled shoulder and elbow angles before contact and its own wrist force during the lift.

E. Controllers

All controllers share the same robot, task and strategies; they differ only in what they know and whether they may decline. *F (fixed rule)*: among strategies completing $\geq 90\%$ of 60 training persons, the one with the lowest median harm. *R (reactive force guard)*: runs F; if the wrist force exceeds m times the 95th percentile of the force in harmless training lifts ($m \in \{0.6, 0.8, 1.0, 1.2\}$), it stops, retracts for 0.5 s and tries the best strategy supporting at least 9 cm away (which loads the other joint), then the next best, up to three attempts. *CAT*, *CAT_{half}*: decline if the (complete or half-documented) categorical flag marks a restriction; otherwise F. *S (record threshold)*: decline if the recorded free range of either joint is below 3.5° ; otherwise F. *M (population body model)*: a library of 199 simulated unrestricted bodies, each run with all 48 strategies, predicts each joint's excursion and each strategy's completion rate. M keeps F if F is predicted to stay at least 3° inside both recorded ranges; otherwise it uses the strategy with the highest predicted completion among those with predicted completion ≥ 0.8 that keep the margin; if none does, it executes the strategy with the smallest predicted overshoot if that is at most 3° , and declines otherwise. *L (policy learned offline in simulation)*: training data are 400 simulated persons from earlier rounds, each with outcomes for all 48 strategies. From six features—the two recorded free ranges (clipped at 30°), two unrestricted flags and the two settled joint angles—a k -nearest-neighbour model ($K = 25$, standardised features) estimates, for each strategy, the probability of a safe completion among the 25 most similar training persons. L picks the strategy with the highest estimate and declines if it is below $\tau = 0.65$, chosen on the training persons as the value with the fewest harmful outcomes among those keeping safe completions within 2.5 percentage points of F. Persons recorded as unrestricted receive F. A variant without posture input (four features, $\tau = 0.60$) was specified in advance. *O (oracle)*: knows the outcome of every strategy and picks a safe

completion if one exists, otherwise declines; a ceiling, not an achievable controller.

F. Protocol and statistics

The study proceeded in confirmation rounds (numbered 17–26). Before each round’s data were generated, we wrote the hypotheses, controllers, thresholds, population, sample size, non-inferiority margins and analysis into a dated protocol document, kept locally and not externally registered; we call hypotheses fixed in this way preregistered. Every confirmation set consists of simulated persons (random seeds) never used before. Deviations included an implementation bug found after round-17 data had been generated, a harm definition corrected after round 22, a change of primary outcome after round 18R, and the reuse of earlier confirmation persons as development and training data (never as confirmation data); details in the supplementary material.

All comparisons are paired on the same simulated persons. We report per-person differences in harmful-outcome rate and safe-completion rate with 95% confidence intervals from a paired t procedure (the preregistered code used $t = 2.1$ for all $n > 20$, which makes the intervals about 6% wider than exact). A harm hypothesis was supported if the upper confidence bound was below zero; for force guards, the preregistered criterion was an upper bound below -5 percentage points. Non-inferiority on safe completions used margins fixed in advance (2–2.5 percentage points); equivalence used ± 0.05 . We did not correct for multiplicity. Analyses marked *post hoc* were not preregistered; their tuned baselines were tuned on the 400 training persons only and evaluated on the 600 confirmation persons of rounds 24 and 25 that have complete outcome grids.

IV. RESULTS

A. Force sensing does not substitute for knowledge

The fixed rule F harmed 34–40% of persons in every hard-limit confirmation set (e.g. 229 of 600 in round 22). Across four guard thresholds, under a hard limit and graded end-feel of 6, 12, 20 and 25°, no force guard reduced the harmful-outcome rate by significantly more than the preregistered 5 percentage points (Fig. 2). This is a failure to show a large reduction, not proof of none: the tightest guard produced small significant reductions at 6° (-4 points), 20° (-4) and 25° (-7 [-11 , -2]), but not with a hard limit or at 12°, and each of these cost 40–51% of the safe completions. With a hard limit, tightening the guard reduced safe completions from 69 to 40 of 120 persons at an unchanged harm rate. The mechanism was timing: at the primary threshold, 95–100% of harmful outcomes occurred during the first attempt, before the controller could switch strategy. It was not a lack of observability: with 12° end-feel, the wrist force tracked joint load within a person (median Spearman correlation 0.75; 0.10–0.35 with a hard limit or 6°), yet the guard still acted after the joint had been loaded.

TABLE I
INFORMATION LADDER: 300 NEW PERSONS, HARMFUL ≥ 0.5 N·M
(ROUND 24)

Knowledge of the person	Safe	Harmful	Declined
None (fixed rule F)	177	110	0
Flag, half documented $\rightarrow$ decline	140	60	88
Flag, complete $\rightarrow$ decline	112	0	179
Ranges $\pm 20^\circ$ + record threshold S	155	51	83
Ranges $\pm 20^\circ$ + body model M	162	67	59
Ranges $\pm 20^\circ$ + learned policy L	174	25	91
Ranges $\pm 4^\circ$ + M / L	176 / 184	64 / 20	49 / 85
Perfect knowledge O	237	0	63

B. The information ladder

The body-model rule M, which declines when it predicts no safe lift, reduced the harmful-outcome rate by 14–26 percentage points in every confirmation condition in which it was tested (rounds 21–26; e.g. -0.19 [-0.23 , -0.16], 113 vs. 229 of 600) with safe completions comparable to F (351 vs. 344). Table I and Fig. 3 give harmful outcomes and safe completions for 300 new persons at each level of knowledge (round 24). A complete categorical flag used to decline removed all harm but also 37% of safe completions; with half the flags missing it left 60 harmful outcomes. Joint ranges with 20° error combined with S or M reduced harm by 39–54%. The learned policy L reduced harmful outcomes to 25, fewer than M (-0.14 [-0.19 , -0.09]), with more safe completions than M ($+0.04$ [0.00 , $+0.08$]) and similar safe completions to F (174 vs. 177; preregistered non-inferiority not established, difference -0.01 [-0.05 , $+0.03$]). Record accuracy mattered little once L was used (Table I).

The remaining distance to perfect knowledge (237 safe, 0 harmful) had two parts. Of the 63 safe completions L missed, 47 (75%) were unnecessary refusals of persons who had a safe strategy. Of L’s 25 harmful outcomes, 18 were persons for whom no safe strategy existed, whom L should have declined. L thus both over- and under-refused.

In threshold-free terms, the mean robot-caused end-range load per person was 6.73 N·m under F, 3.19 under M and 0.79 under L (L – F: -5.94 [-7.78 , -4.10]; L – M: -2.41 [-3.69 , -1.12]). The advantage is not only a matter of lifting fewer people: post hoc, on the 423 of 600 persons L lifted (rounds 24–25), L harmed 53 and F would have harmed 95 (-0.10 [-0.13 , -0.07]; mean load 1.63 vs. 2.31 N·m).

C. The learned policy under degradations

L was then tested on further new persons (Fig. 4). It replicated its reduction relative to F (-0.27 [-0.32 , -0.21]) and to M (-0.08 [-0.12 , -0.03]); its safe completions relative to M were -0.01 [-0.05 , $+0.03$], so the preregistered non-inferiority to M did not replicate. The reduction relative to F and M held with Gaussian noise (SD 5°) on the observed posture or without posture input, in the shifted population with more frequent and tighter contractures (170 vs. 38 harmful of 250, without retraining), and under 20° graded end-feel. In

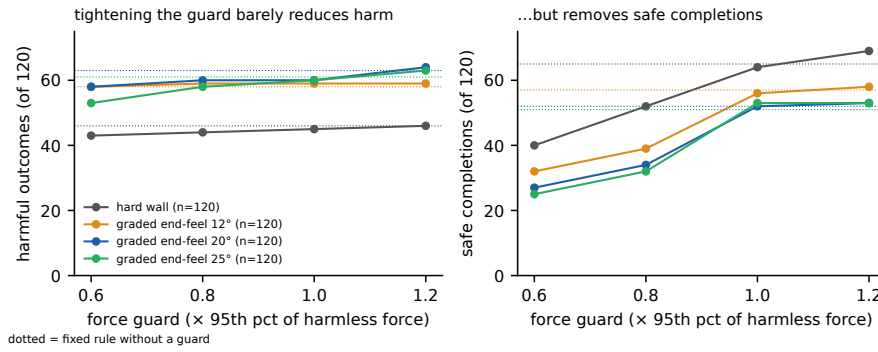


Fig. 2. Force-threshold control at four thresholds: hard limit (round 21, 120 new persons) and graded end-feel of 12, 20 and 25° (round 23, the same 120 new persons under each width). Tightening the guard barely changes harmful outcomes (left) but removes safe completions (right).

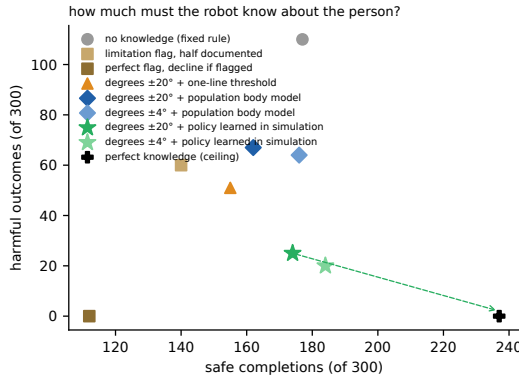


Fig. 3. Information ladder on 300 new persons (lower right is better). The dashed arrow is the remaining gap from the learned policy to perfect knowledge.

these conditions its safe-completion rate was never more than 1.3 percentage points below F's.

Record completeness mattered more. With half of the restrictions missing from the record, L still harmed fewer persons than F (64 vs. 108) and M (81), but its harm was 59% of F's rather than about a quarter, and a categorical flag used to decline (CAT_{half}) harmed fewer persons than L (50 vs. 64; +0.05 [+0.02, +0.07]) at 29 fewer safe completions. By construction, a record-based controller treats a person whose restriction is not recorded like the fixed rule: the 85 such persons were harmed at exactly the fixed-rule rate (50 vs. 50).

We also tested L, M and F unchanged in two worlds whose physics differed from the training simulator: three-fold stretch-reflex gains and wider individual variation ($\sigma = 0.40$). L's reduction relative to F and M held in both, with safe completions 3.3 and 0.3 percentage points below F's. These perturbations left the joint-limit geometry unchanged, and F's harm rate barely moved (36% and 40%, vs. 36–37% in the standard world), so they test robustness to muscle dynamics and body variation rather than to a different body structure.

Three preregistered checks used the 600 confirmation persons with complete outcome grids. *Is F a straw man?* In hindsight, the best of all 48 fixed strategies harmed 203

persons, but with only 135 safe completions; the preregistered F harmed 218 (rank 9 of 48); L harmed 53. *Does L shift load?* On the persons L lifted, glenohumeral shear did not increase (paired mean difference L – F –3.6 N, median 0). *Stale records:* when records over-estimated the free range of restricted joints by 5° or 10°, L's harmful outcomes rose from 53 to 57 and 68, whereas M's rose from 118 to 160 and 210.

At the secondary 2.5 N·m threshold (preregistered), L reduced harm in every confirmation set but completed 4–8 percentage points fewer lifts than F; in a post hoc threshold sweep (round 24), F completed more lifts safely than L at almost every threshold (e.g. 203 vs. 186 at 2.5 N·m, harming 80 vs. 12).

D. What the robot needs to know (post hoc)

On the 600 confirmation persons with complete grids, we compared L with baselines tuned on the 400 training persons only (Fig. 5). *Which joint, not how many degrees:* a nearest-neighbour policy given only the two restriction flags and the settled posture ($\tau = 0.65$ selected by the same criterion) performed like L: 353 safe and 46 harmful vs. 341 and 53 (L – flags: harmful +0.01 [–0.01, +0.03], safe –0.02 [–0.04, 0.00]). *A lookup table is coarser:* the best training strategy per restriction type (none, shoulder, elbow, both) gave only four operating points; at the tuned setting it completed more lifts (396) at a higher harm (94). *Re-tuning the rules does not close the gap:* re-tuned for 20° records, M harmed 113 persons at 335 safe completions (L – M: –0.10 [–0.13, –0.07]); S could not meet the completion criterion at any setting and reverted to never declining. Across the whole trade-off, the learned policies lay below the record-threshold and body-model curves. By restriction type, L lifted every unrestricted person with F (224 safe, 0 harmful), declined everyone with both joints restricted (71, of whom only 17 had any safe strategy), and for single-joint restrictions reduced harmful outcomes relative to F from 75 to 30 (shoulder) and from 82 to 23 (elbow).

E. What did not help

Routing motion to a joint with spare range by choosing the support site, without declining, did not reduce harm in

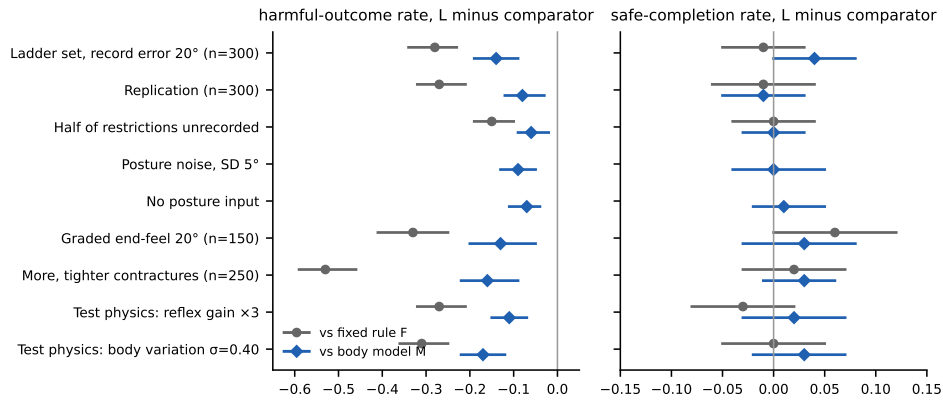


Fig. 4. Robustness of the learned policy L: paired differences (95% confidence intervals) in harmful-outcome rate (left) and safe-completion rate (right) between L and F (grey) or M (blue), in every preregistered confirmation condition (harmful ≥ 0.5 N·m). No controller was retrained; the no-posture variant is the separately specified four-feature policy; posture conditions were compared with M only, as preregistered (posture noise: Gaussian, SD 5°).

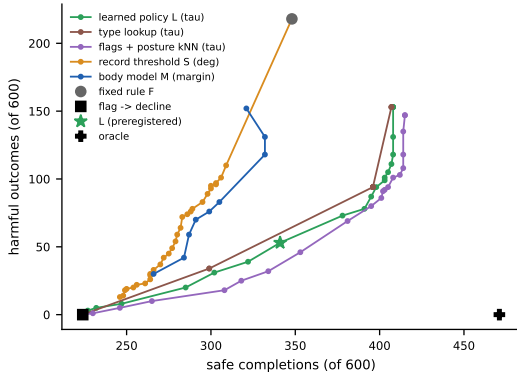


Fig. 5. Post hoc trade-off between safe completions and harmful outcomes on the 600 confirmation persons with complete grids (rounds 24 and 25), sweeping each rule's decline parameter. Lower right is better.

round 21 (-0.03 [-0.09 , $+0.02$]; preregistered null); M's effect came mostly from declining. Identifying the individual body by two gentle probe lifts was supported in round 18 ($n = 40$) but did not replicate in a frozen replication with 120 persons; in round 19 the probes themselves loaded a restricted joint beyond 0.5 N·m in 18 of 71 probed persons, and the probing controller harmed more persons than the same rule without probing (29 vs. 21 of 120). In round 19 the body model with declining (66 safe, 21 harmful of 120) reduced harm relative to the reactive guard (76 safe, 42 harmful) but did not match its safe completions; the learned policy was not compared with the guard directly.

V. DISCUSSION

Why knowledge, and not force. In end-effector support of a multi-joint limb, the loaded joint is not the one the force sensor sees, and the load builds within the first attempt; a guard can only react after the joint has been loaded. Force limits remain necessary for skin and soft tissue, but they are not sufficient for joints. We did not test a kinematic guard

that stops when the observed joint excursion approaches the recorded range.

What kind of knowledge. The robot did not need a detailed personal model. What mattered was knowing *which* joint is restricted, and having learned, from the simulated outcomes of many persons like this one, which lifts are safe and when to decline. M's library contains only unrestricted bodies and must infer harm from predicted excursions, whereas L learns harm labels directly from simulated persons of the restricted population, and kept its advantage in a more impaired population than it was trained on. The value of the simulator as a world model lay in generating outcome labels from which a decision policy can be learned offline.

The cost of declining. Declining is not free: it hands the task to a human carer, whose handling carries its own risk, or delays it. L declined 30% of persons, the oracle 21%. Because τ was chosen to keep safe completions close to F's, similar completion counts are partly a design choice; a deployed system would choose τ with the care team. For human carers, evidence that more detailed assessment reduces handling injury is lacking [34], [35], so our ladder describes robots, not carers. Our simulated persons do not wince or speak; the results apply most directly to people who cannot give such cues reliably.

Implications for design. Robots that move limbs by end-effector contact should (i) consult joint-level information before contact—at least which joints are restricted; (ii) include an explicit option to decline and hand over the task; and (iii) not rely on force thresholds to protect joints. Because unrecorded restrictions cannot be protected, the completeness of the record matters more than its precision.

Limitations. All results are simulated; no robot or human data were collected. Training and test simulators share the same musculoskeletal structure and joint-limit geometry; a real person differs in more ways, including voluntary resistance, pain responses, soft-tissue contact and body geometry. The harm measure is a mechanical exposure proxy without a pain or injury dose-response and depends on the simulator's

constraint model. The independence of shoulder and elbow restriction and the uniform free ranges are assumed. We studied one task, one end effector and a passive recipient, and compared only a force-based reactive controller. The protocol was dated and kept locally rather than registered externally, earlier confirmation persons were reused for development, and the analyses of what the robot needs to know are post hoc.

VI. CONCLUSION

For a robot lifting a person's arm, gentleness depended less on sensing the contact than on knowing the person before contact—above all, which joint is restricted and when not to lift. A policy learned offline in simulation turned a coarse, error-prone care record into about a quarter of the harm of a fixed lift at a similar number of completed tasks. Force thresholds could not substitute for that knowledge, and restrictions missing from the record remained unprotected.

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